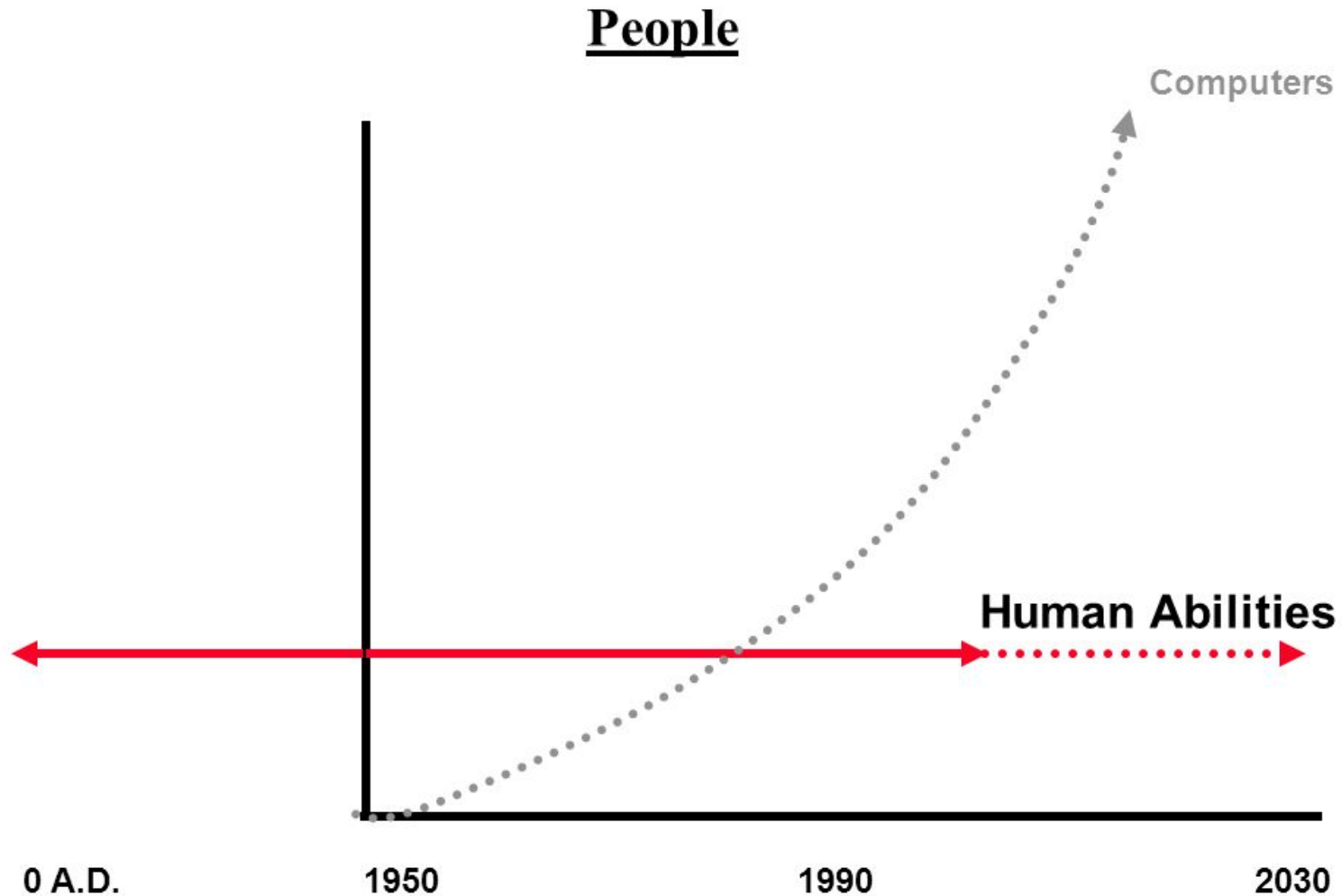


Background

- Increasing use of automated systems
 - Hardware and software technology are improving rapidly
 - User interface technology is lagging
- Critical bottleneck is now user interface, not processing
- Two powerful information processors communicating via narrow-bandwidth, highly constrained interface
 - Goal: Increase bandwidth across channel
 - Start from human's perceptual abilities, processes, organs
 - Then work toward devices, interaction techniques, not reverse

Human Abilities vs. Computer Abilities



These graphs are variations of those presented by Bill Buxton

Computer Science and HCI

- Problems and opportunities in this area increasing, not decreasing, with new technology
 - Easier to use
 - Harder to program
- Simple engineering methods for good interfaces do not exist yet
 - Still need iterative development, prototyping, and testing
 - Hence user interface code is most changeable part of a system

What is a human-computer interface?

- Definition of all inputs from user to computer
 - Definition of all outputs from computer to user
 - Definition of sequencing of inputs and outputs
-
- **What** the user sees/perceives
 - **Not how** it is implemented

The User Interface Is Important

- User interface strongly affects perception of software
 - Usable software sells better
 - Unusable web sites are abandoned
- Perception is sometimes superficial
 - Users blame themselves for UI failings
 - People who make buying decisions are not always end-users

The Cost of Getting It Wrong

- Users' time isn't getting cheaper
- Design it correctly now, or pay for it later

User Interfaces Are Hard to Design

- You are not the user
 - Most software engineering is about communicating with other programmers
 - UI is about communicating with users
- The user is always right
 - Consistent problems are the system's fault
- ...but the user is not always right, either
 - Users aren't designers

Usability Defined

- Usability: how well users can use the system's functionality
- Dimensions of usability
 - Learnability: is it easy to learn?
 - Efficiency: once learned, is it fast to use?
 - Visibility: is the state of the system visible?
 - Errors: are errors few and recoverable?
 - Satisfaction: is it enjoyable to use?

Usability Dimensions Vary In Importance

- Depends on the user
 - Novice users need learnability
 - Infrequent users need memorability
 - Experts need efficiency
- But no user is uniformly novice or expert
 - Domain experience
 - Application experience
 - Feature experience

User Experience Design

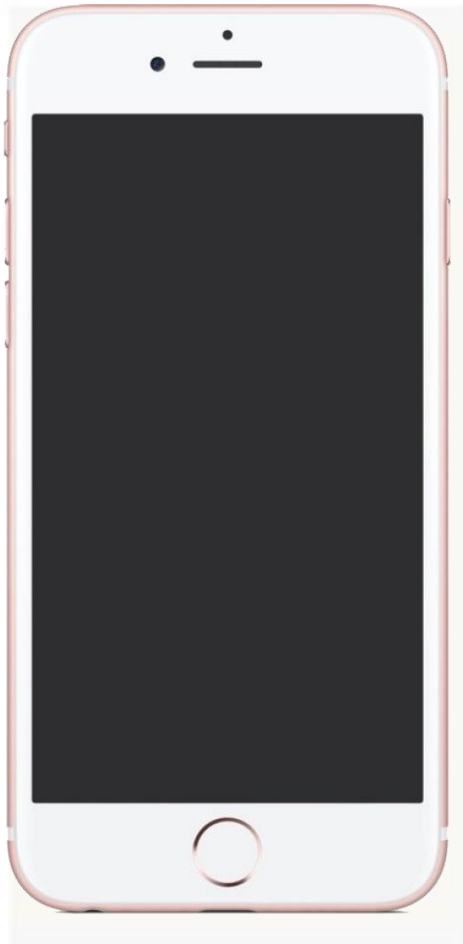


Image courtesy of [YU-TA LEE](#) on Flickr.

Usability Is Only One Attribute of a System

- Software designers have a lot to worry about:
 - Functionality
 - Performance
 - Cost
 - Security
 - Usability
 - Size
 - Reliability
 - Standards
- Many design decisions involve tradeoffs among different attributes